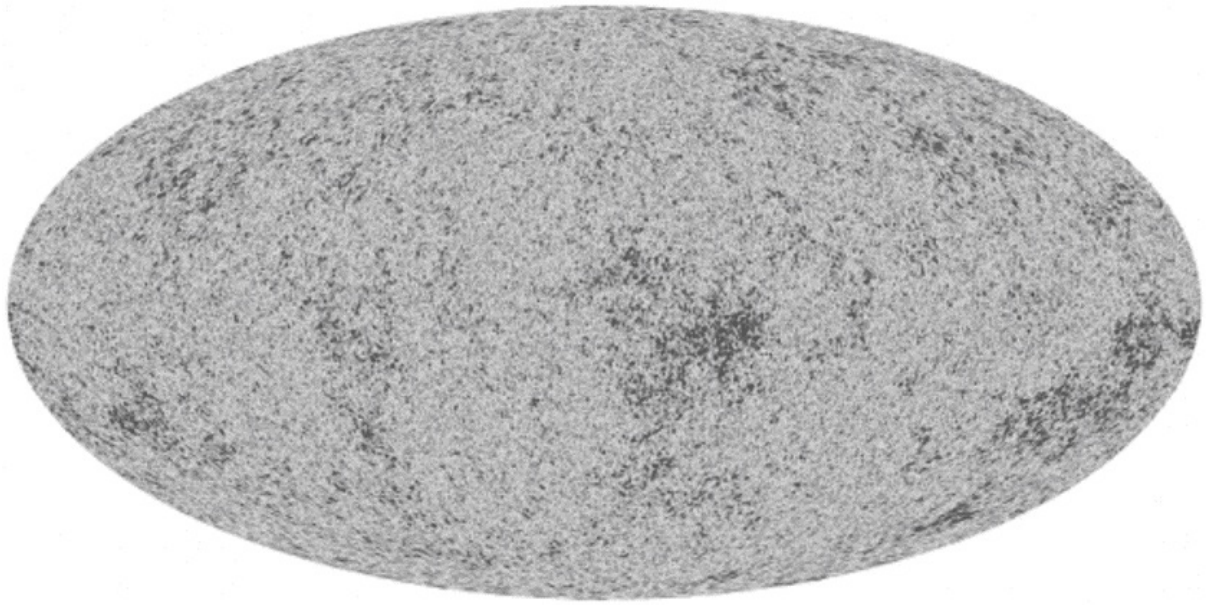


Figure 9.3 The fluctuations of the cosmic background radiation. This is the image of the oldest object in the universe available to us. These fluctuations were produced 14 billion years ago. In the statistics of such fluctuations we hope to find confirmation of the predictions of quantum gravity.

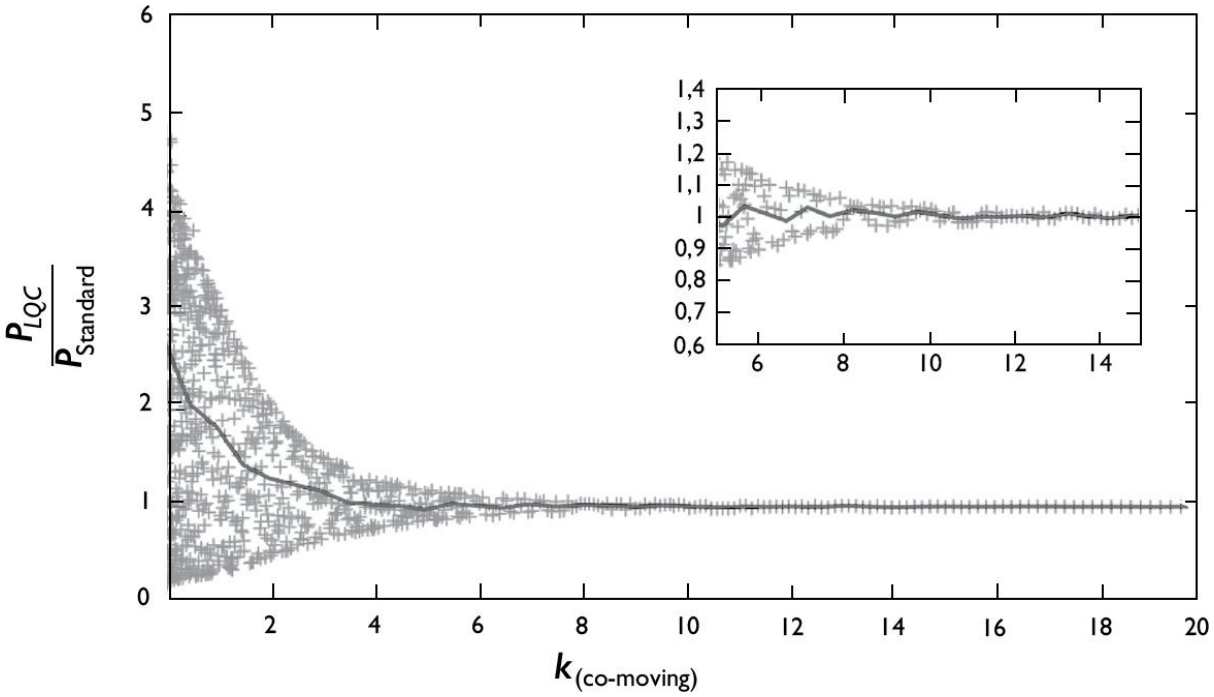


Figure 9.4 Possible prediction of the spectrum of background radiation, of loop quantum gravity (shown by the solid line) compared with the current experimental errors (as represented by the points). Courtesy of A. Ashtekar, I. Agullo and W. Nelson.

Traces of the great primordial heat must also be in the gravitational field itself. The gravitational field, too, that is to say, space itself, must be tremulous like the surface of the sea. Therefore, a cosmic *gravitational* background radiation must also exist – older even than the electromagnetic